From Captive to Contestable: When Platforms Compete, Who Pays?

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June 30, 2026

Abstract

When platforms compete next to sellers' offline storefronts, who pays? We study this question using China's 2025 food-delivery subsidy war between Meituan and Ele.me. Merchant-level data spanning all channels reveal a *traffic–revenue paradox*: subsidized delivery orders surged yet total revenue fell, as online orders cannibalized high-margin dine-in sales. A structural model of competing platforms with an endogenous offline outside option shows consumer subsidies act as a tax on the storefront channel. Calibration reveals the war is overwhelmingly redistribution—from merchants and platforms to consumers—with negligible net surplus gains. Commission caps alone can defuse the subsidy spiral.

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1 Introduction

In early 2025, a massive food-delivery subsidy war erupted across China. The two leading platforms, Meituan and Ele.me, began subsidizing consumers aggressively—channeling massive discounts into the delivery market and pushing daily orders past 200 million. Regulators responded with immediate concern, summoning the platforms within weeks to demand an end to what they termed "involution-style" competition. The episode raises a question the standard framework leaves open: when platforms compete, who pays—the consumers and platforms inside the contest, or the sellers who also trade offline?

The 2025 subsidy war provides an ideal laboratory to address this question, offering two distinct features.

First, the setting is unique because the platform and the merchant's own offline channel compete head-to-head. In standard two-sided market models, the platform operates in isolation, and the seller's outside option is treated as an exogenous payoff. In our setting, however, the merchant's outside option is a physical storefront. When platforms compete, they do not just fight each other; they compete directly with the storefront for the same pool of demand. Every online order that the platform wins directly cannibalizes the merchant's high-margin dine-in business. We thus analyze how platform competition endogenously erodes the seller's offline channel—a novel margin to standard models.

Second, the regulatory response was remarkably swift and aggressive, highlighting the urgent need for policy evaluation. Within weeks of the outbreak, the State Administration for Market Regulation (SAMR) demanded an end to "involution-style" competition, culminating in the June 2026 draft of the "Ten Guidelines on Subsidy Practices" to explicitly prohibit consumer subsidies. This aggressive regulatory push makes it critical to understand the welfare and distributional consequences of capping platform competition. Because these policies are newly proposed, evaluating their effects requires a structural model—allowing us to analyze the very interventions regulators are poised to deploy.

We ask two questions. First, when platforms compete in the presence of a direct selling channel, who ultimately bears the cost? Second, does platform competition still improve welfare once we account for the offline market it cannibalizes? The answers matter for policy. Our key message is not simply that platform competition has spillovers. Rather, it is that when platforms compete, the outside market becomes a policy-relevant margin—one that standard competition analysis does not model; cannot measure from contract data; and cannot directly regulate.

Based on public indicators, the immediate winners and losers of the subsidy war are clear: consumers enjoyed windfall discounts, while the platforms bled cash.[1] Yet, the impact on the

[1]By their own 2025 quarterly earnings reports, the competing platforms together absorbed over CNY 100 billion

merchants who actually supply this market remains unclear. To pin down these effects, we leverage a unique and highly granular dataset from a national restaurant payment platform. The data record, at the merchant-day level, orders and revenue across all of a restaurant's channels—dine-in as well as delivery—together with both posted and post-subsidy prices. Using a difference-in-differences design that compares the same merchants across the 2025 war window and the matched 2024 window, we document a *traffic–revenue paradox*. The war did what subsidies are meant to do on the delivery channel: it cut delivery prices and raised delivery orders. But the extra traffic did not translate into higher profits. Total orders rose only modestly while total merchant revenue *fell*, because subsidized delivery pulled customers away from the high-margin dine-in counter.[2] A triple-difference design using dine-in-only restaurants as a control confirms that this contraction is concentrated in merchants exposed to within-store channel substitution—the signature of our proposed mechanism rather than of a general economic downturn.

We develop a model of competing platforms with direct selling channls to explain why traffic and revenue move apart, and to turn the empirical pattern into a statement about welfare. In the model, a continuum of local merchants sell through both an online platform and their own storefronts, while consumers choose where and from whom to buy. The central object of the analysis is the storefront markup. We show that this markup is determined by the ratio of the channel-substitution parameter to the online share of trade. As the platforms subsidize consumers and pull them online, they systematically drain the offline channel. The remaining offline customer pool becomes thinner and more price-sensitive, which erodes the storefront's pricing power. A consumer subsidy is therefore, in effect, a tax on the merchant's outside option. The resulting harm is "off the books": commissions and posted prices can remain unchanged while merchant profits collapse, because the loss runs through the erosion of the offline markup.

We characterize the war itself as a structural shift from a *captive* duopoly to a *contestable* one. Before the war, most consumers used a single app and rarely compared platforms; the two platforms were linked only through the shared offline channel they both eroded. The war made consumers willing to search across apps. We show that this single structural change has sharp, unconditional consequences. Moving from captive to contestable competition raises online penetration, expands the merchant base, and drives up both commissions and subsidies. In doing so, it compresses combined platform profits—rendering the war a prisoner's dilemma the platforms inflict on themselves. While the shift raises consumer surplus, it lowers merchant surplus: the collapse of the offline markup dominates the modest gain merchants receive from a larger online

(USD 14 billion) in subsidies, with Meituan reporting its largest operating loss since listing and Alibaba's profit collapsing.

[2] Specifically, the subsidy war cut average delivery prices by about 15 percent and raised delivery orders by about 16 percent. However, total merchant revenue fell by about 7 percent on net, driven by a 12 percent contraction in high-margin dine-in orders.

business. Crucially, these qualitative results hold for any reasonable distribution of merchant costs, independent of the calibration.

A calibration to the 2025 episode puts magnitudes on these changes, supplying the welfare arithmetic that theory alone cannot pin down. We find that the war is overwhelmingly a redistribution: consumers gain about 9.3 yuan per order, while merchants lose about 7.4 and platforms lose about 1.2, funded by the collapse of the offline markup. The net effect on total surplus is small and second-order; the distributional story is first-order and robust.

Counterfactual analyses then evaluate the policies regulators actually deployed. Capping the subsidy directly moves the market toward the efficient allocation, but it transfers surplus from consumers back to platforms and merchants. More striking, capping the merchant commission alone is sufficient to defuse the war. Once platforms can extract less from each online order, their privately optimal subsidy falls endogenously, returning to roughly the pre-war level. A "fee-relief" rule thus carries an "anti-subsidy" effect for free. Crucially, our analysis reveals a fundamental distributional impossibility result for the regulator: because merchants and platforms always move together and opposite to consumers, no single policy instrument can protect sellers and shoppers at the same time.

This paper makes three contributions. First, we document a *traffic–revenue paradox* using granular merchant-level data, providing direct evidence on how platform competition affects the sellers who supply the market. Second, we explain this paradox through a tractable model of platform competition featuring an endogenous merchant outside option. Third, we calibrate the model to evaluate regulatory interventions, showing that capping commissions can defuse a subsidy war through a waterbed effect.

Related literature.

Online platforms and the offline channel. A growing literature documents how digital platforms reshape the offline businesses of sellers beside them. Airbnb's entry cuts hotel revenues, hitting low-price hotels hardest (Zervas et al., 2017), with consumer gains concentrated where hotel capacity binds and total surplus rising only modestly (Farronato and Fradkin, 2022); Amazon fulfillment centers reduce nearby store sales and accelerate store exit (Chava et al., 2022); and online–offline substitution is by now a robust regularity across retail, media, and advertising (Goldfarb and Tucker, 2019). A contrasting result is Li and Wang (2025): the *entry* of delivery platforms *raises* restaurant revenue, with positive dine-in spillovers, through demand discovery—delivery reaches consumers who would not otherwise have visited in person.

Our setting differs in two major ways. First, the incumbents in previous studies usually do not sell through the platform—hotels do not list on Airbnb, retailers do not sell through Amazon—so the harm is demand loss to an outside rival; here the same restaurant runs both its dine-in counter and its delivery business, and the mechanism is channel substitution *within* the merchant's own operation. Second, those papers study platform *entry*, whereas our shock is platform *competition*: a subsidy war between two incumbents over a fixed local consumer pool. When the platforms already serve the market, the demand-discovery channel is absent—every consumer already knows the restaurant—so the war adds no customers but diverts them from the high-margin counter to the lower-margin delivery channel, while rising commissions squeeze online margins. The result is our traffic–revenue paradox: more orders, less revenue.

Two-sided markets and platform competition. The paper is closely related to the two-sided market literature (Caillaud and Jullien, 2003; Rochet and Tirole, 2003, 2006; Armstrong, 2006), surveyed in Rysman (2009) and Jullien et al. (2021). We build on this framework, extending its standard commission and subsidy formulas. Our key departure is making the seller's outside option endogenous. In standard competitive-bottleneck models, the value of not participating on a platform is a fixed number. In our model, this value—the storefront markup—is determined in equilibrium by online penetration. When platforms compete harder and pull more consumers online, they jointly erode the merchant's offline channel. Because this erosion occurs off the platforms, the resulting harm is invisible on all observable contract terms: commissions and posted prices can remain unchanged while merchant profits collapse. The paper closest to ours in the intermediation literature is Gautier et al. (2023), who study how sellers choose between a direct market and a platform when both are available; we study how the value of the direct market is itself endogenously determined by the intensity of platform competition.

Platform regulation. Our counterfactual analysis connects to the literature on platform fee regulation. Li and Wang (2024) show that capping commissions in US cities reduced welfare and harmed independent restaurants, as platforms strategically redirected demand toward chains. While their study highlights how regulating one platform instrument triggers evasive adjustments on others, we document a strategic link working in the regulator's favor. Because commissions and subsidies are jointly determined, capping the commission endogenously reduces the platform's optimal subsidy. A fee-relief rule thus carries an anti-subsidy effect for free, allowing regulators to defuse the war without a direct subsidy ban.

Roadmap. Section 2 describes the subsidy war. Section 3 presents the data and the empirical results. Section 4 develops the model and its comparative statics. Section 5 calibrates the model and reports the welfare and counterfactual analysis. Section 6 concludes.

2 Institutional Background

2.1 The 2025 Subsidy War

China's food-delivery market is among the largest in the world. In 2024 it served roughly 590 million users, more than half of the country's internet population, through over ten million merchants, intermediating on the order of CNY 1.6 trillion in gross merchandise value. Because every delivery transaction originates at a physical restaurant, the platforms' competitive conduct is transmitted directly to a vast population of brick-and-mortar merchants. Until 2025 the market was a stable and highly concentrated duopoly: Meituan held roughly 65 percent of orders and Alibaba's Ele.me roughly 33 percent, with all remaining platforms together accounting for only about 2 percent, a configuration that had persisted with little change for years.[3] Just as important, by 2024 the market had matured. Annual gross-merchandise-value growth had decelerated sharply, from about 37 percent in 2023 to roughly 7 percent in 2024, and delivery penetration had plateaued; the sector had entered what industry observers term a phase of "stock competition," in which additional orders for any platform increasingly had to be won from rivals, or from other channels, rather than from an expanding pool of new demand. This backdrop bears on how we read the war's effects: in a saturated market, aggressive subsidization redistributes existing transactions at least as much as it generates new ones.

The 2025 price war erupted against this backdrop and escalated in three stages. In February 2025, JD.com entered food delivery, courting sit-down restaurants with zero-commission onboarding and pledging full social-insurance coverage for its couriers, backed by a widely publicized "ten-billion subsidy" program. For three months this remained a one-sided challenge to the incumbents. The decisive escalation came on April 30, 2025, when Alibaba entered in force on two fronts at once: Ele.me launched a platform-wide "over-ten-billion" subsidy campaign, while Alibaba's instant-retail arm, rebranded Taobao Flash Purchase, opened a top-level traffic entry on the Taobao app and pooled its subsidies with Ele.me's merchant supply. Meituan, the incumbent leader, responded defensively in kind, pledging CNY 100 billion over three years. A one-on-one contest thus became a three-way subsidy war among Meituan, Alibaba, and JD.com. By July 2025 the competition had intensified into free orders and zero-yuan deals, the most aggressive consumer pricing in the sector's history.[4] We date the war to the April 30 escalation throughout.

[3]The duopoly dates to 2018, when Ele.me absorbed Baidu's delivery business. Meituan's lead was also one of profitability: industry estimates place its delivery operating profit near CNY 32 billion in 2024, against roughly break-even for Ele.me, whose parent reported an adjusted EBITA loss of about CNY 9.8 billion in its local-services segment that fiscal year.

[4]Subsidized order volumes escalated rapidly: Taobao Flash Purchase reported 10 million orders per day by May 5 and 40 million by late May, and by early July combined daily orders across the three platforms peaked near 250 million, more than double the 2024 level. Alibaba's broader aim was the instant-retail market ("everything delivered in thirty minutes"), for which restaurant delivery served as the entry point.

2.2 Consumer Search and Multi-Homing

A defining feature of the 2025 subsidy war was the drastic shift in consumer search behavior. Prior to the war, the Chinese food-delivery market was characterized by high consumer loyalty and low cross-platform search. Industry reports from mobile data tracking firms (such as QuestMobile and Analysys) show that before 2025, the active user overlap between Meituan and Ele.me was remarkably low, hovering around 18 percent. Most consumers were effectively "captive" to a single platform, driven by habitual use and deep integration within parent ecosystems—specifically, Tencent's WeChat for Meituan and Alibaba's Alipay for Ele.me.

The outbreak of the subsidy war in April 2025 shattered this captivity. The introduction of massive, platform-financed discounts—such as Meituan's "Shenquan" coupons and Ele.me's daily price-matching subsidies—made the price differences between the two platforms highly salient. In response, consumers became highly active searchers, downloading both apps to compare prices before placing an order. QuestMobile data from the peak of the war in late 2025 reveal that the monthly active user overlap between Meituan and Ele.me surged to over 46 percent. Rather than remaining loyal to a single platform, a large portion of the consumer base became "contestable," toggling between apps to exploit the highest available subsidy on any given day. This transition from captive to contestable consumer search forms the empirical foundation for the market structure shift we model in Section 4.

2.3 Empirical Identification and Margin Compression

Two features of this episode are central to our identification strategy, and a third to interpreting its incidence. First, the April 30 escalation was a sharply dated, sector-wide event driven by competition *among the platforms*, not by conditions at any individual restaurant. Second, individual merchants were price-takers in this game: the subsidies were designed and financed platform-side and imposed on the merchant population at large, leaving little scope for anticipatory or strategic response at the restaurant level. Third, the economics of the platform relationship meant that this conduct bore directly on merchant margins. Even before the war, platforms levied commissions of roughly 6 to 8 percent of order value, on top of delivery and promotion fees, margins thin enough that JD.com's zero-commission pitch could serve as its wedge into the market. When the subsidy war compressed the prices consumers paid without a commensurate change in these fees, much of its incidence fell on the merchant side of the market rather than on the platforms that set it. We exploit the first two features for identification in Section 3.2 and return to the third when we interpret the revenue results in Section 3.3.

Regulatory Response The regulatory response from the Chinese government was remarkably swift, escalating from initial administrative warnings to binding prohibitions with unusual speed. The primary objective of these interventions was clear: to dismantle the subsidy war and stamp out what authorities termed "involution-style" competition. In May and July 2025—mere weeks after the war's outbreak—the State Administration for Market Regulation (SAMR) summoned the platforms twice to demand an immediate halt to their predatory pricing campaigns. As the platforms dragged their feet, the regulatory machinery accelerated: in December 2025, a national standard for Delivery Platform Services was released; in January 2026, the State Council launched a formal anti-monopoly investigation; and by June 2026, SAMR had drafted binding rules—the "Ten Guidelines on Subsidy Practices"—to explicitly outlaw consumer subsidies.

Because this regulatory intervention is so explicit and aggressive, evaluating its welfare and distributional consequences is highly necessary. Since these policies are newly drafted, they cannot be evaluated using post-policy data. This makes a structural model essential. We use our model to perform this evaluation in Section 5.3, assessing the impact of the very subsidy caps and regulations the authorities have proposed.

Consumers and Platforms On the consumer side, the subsidy war was felt as a windfall. Industry survey data reveal that the massive discounts prompted widespread adoption; average out-of-pocket spending per order fell by over 20 percent; and consumer satisfaction with platform promotions reached record highs. For the platforms, however, this surge in traffic came at a steep price. Financial disclosures from Meituan and Alibaba (Ele.me) paint a clear picture of profitless growth: while daily order volumes reached historic peaks, marketing expenses ballooned, severely compressing operating margins and driving Ele.me's local-services losses to new depths. The platforms were trapped in a prisoner's dilemma.

Yet, this focus on consumers and platforms leaves the third—and arguably most vulnerable—group of actors in the dark: the merchants. How did the subsidy war affect the millions of brick-and-mortar restaurants that form the backbone of the delivery ecosystem? While consumer surveys and platform earnings reports are public and widely scrutinized, they say nothing about the micro-level impact on individual shopfronts. To pin down these effects, we leverage a unique and granular merchant-level dataset from a national restaurant payment provider. We describe this dataset, and the empirical setting it reveals, in Section 3.1.

3 Empirical Evidence

This section presents our empirical analysis of the subsidy war's effect on restaurants. We describe the data and the raw patterns (Section 3.1), lay out the research design (Section 3.2), report the main treatment effects (Section 3.3), and isolate the channel-substitution mechanism behind them (Section 3.4). The evidence is deliberately focused on the merchant side, the party whose fortunes in the war are least visible and least understood.

3.1 Data and Descriptive Evidence

3.1.1 Data

Our data come from a national restaurant payment platform that covers China's mainstream payment scenarios, with a market share approaching 20 percent. The platform records, at the merchant-day level, the orders and revenues a restaurant transacts across *all* of its channels, dine-in as well as delivery through every major platform (Meituan, Ele.me, and JD), which lets us observe channel substitution within the same merchant rather than inferring it across data sources. For each transaction we observe both the *listed* value (revenue at posted menu prices) and the *actual* value (post-subsidy receipts), a dual-price structure that is essential for measuring how the incidence of the subsidies is split between consumers and merchants.

We construct a balanced panel of merchants observed at daily frequency over the March–September window of both 2024 and 2025.[5] To focus on continuing incumbents and to ensure comparability across years, we retain merchants that operate on more than 200 days in each sample year. The resulting panel comprises more than 40,000 restaurant merchants and broadly covers 359 prefecture-level cities and provincially-administered counties nationwide, with approximately 19.3 million merchant-day observations. The treatment window, the post-April-30 period running through September 30, and the matched 2024 calendar window form the basis of the difference-in-differences design developed in the next section.

Table 1 reports summary statistics for the pooled sample. The average restaurant transacts about 140 orders and CNY 6,093 in actual revenue per day, split between a dine-in operation (79 orders) and a delivery operation (61 orders). Two patterns foreshadow the analysis. First, actual revenue (mean CNY 6,094) runs well below listed revenue (mean CNY 6,950), reflecting the discounts already embedded in the market even before the war; the gap between these two series is exactly what the subsidy war widens. Second, the dine-in channel is the more valuable one per transaction: the average dine-in order is worth CNY 71.0 against CNY 37.1 for the average delivery order, so that dine-in accounts for a larger share of revenue (61.4 percent) than of orders (56.0 percent). A shift of volume from dine-in toward delivery therefore mechanically erodes revenue even at unchanged prices, the seed of the traffic–revenue paradox we document below.

[5]The window begins in March by design: we exclude January and February because the lunar New Year falls on different Gregorian dates across years, and the associated return migration generates extreme, year-specific seasonality in restaurant activity. Dropping these two months removes a source of year-over-year non-comparability that the fixed effects alone could not fully absorb.

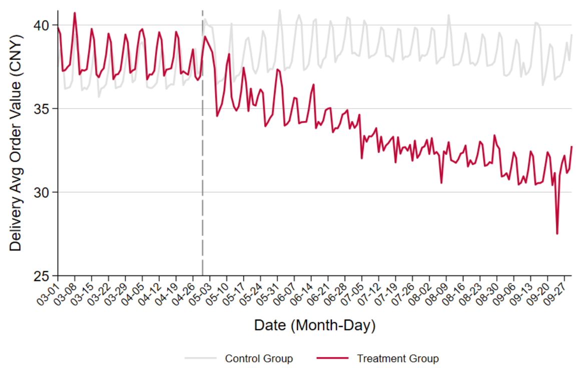
Table 1: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Combined activity					
Total daily orders	19,310,966	140.4	120.9	0	5,399
Total listed revenue (CNY)	19,310,966	6,950.4	8,811.8	0	2,028,691
Total actual revenue (CNY)	19,310,966	6,093.7	8,195.4	0	700,784.7
Dine-in channel					
Dine-in orders	19,310,966	79.2	88.7	0	3,634
Dine-in listed revenue (CNY)	19,310,966	4,366.5	7,477.4	0	2,020,671
Dine-in actual revenue (CNY)	19,310,966	4,253.9	7,196.3	0	700,681.0
Dine-in actual order value (CNY)	18,670,638	71.0	98.1	1.0	29,470.8
Dine-in listed order value (CNY)	18,670,638	73.0	101.9	1.0	37,127.2
Delivery channel					
Delivery orders	19,310,966	61.2	68.5	0	5,311
Delivery listed revenue (CNY)	19,310,966	2,583.9	2,970.3	0	214,444.7
Delivery actual revenue (CNY)	19,310,966	1,839.8	1,991.7	0	213,973.5
Delivery actual order value (CNY)	18,670,638	37.1	27.1	1.0	1,312.7
Delivery listed order value (CNY)	18,670,638	47.9	30.7	2.0	2,412.3
Channel composition					
Dine-in order share (%)	18,670,638	56.0	23.0	0.1	99.9
Dine-in revenue share (%)	18,670,638	61.4	22.8	0.0	100.0

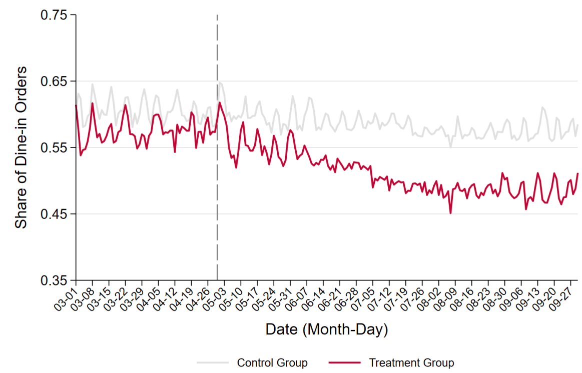
Notes. Merchant-day observations pooled over the March–September windows of 2024 and 2025 for the balanced panel of more than 40,000 merchants across 359 cities. “Listed” values are at posted menu prices; “actual” values are post-subsidy receipts. Order-value and channel-share variables are defined on merchant-days with positive activity, yielding a smaller observation count.

3.1.2 Stylized Facts

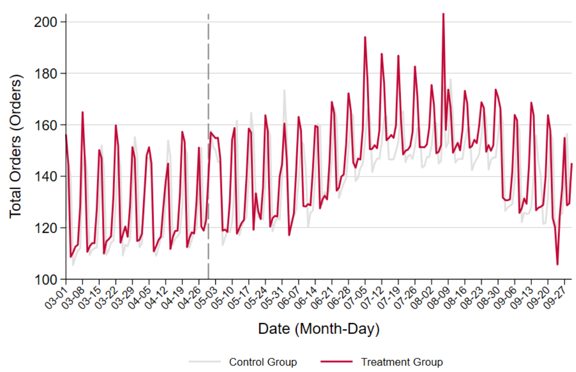
Figure 1 previews the main results graphically. Each panel plots a daily series for the treatment year 2025 (red) against the same calendar window in the control year 2024 (gray), with the vertical dashed line marking the April 30 escalation. The two years track each other closely before April 30 and diverge sharply afterward, a visual counterpart to the parallel-trends logic of our design. Panel (a) shows the average delivery order value collapsing in 2025 once the war begins. Panel (b) shows the dine-in share of orders falling below its 2024 path, as subsidized delivery draws volume away from the counter. Panels (c) and (d) display the central divergence: total orders in 2025 rise above the 2024 path while total revenue falls below it. Traffic and revenue move in opposite directions, precisely the paradox we quantify in Section 3.3.



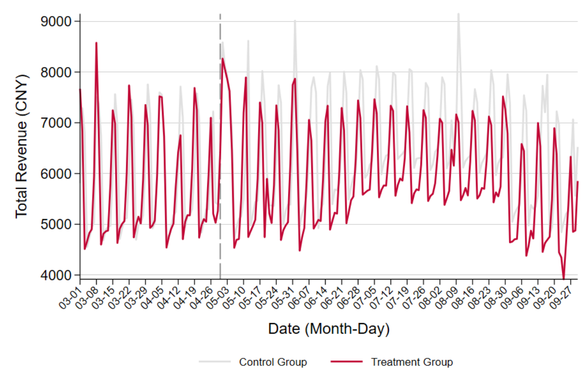
(a) Delivery average order value



(b) Dine-in order share



(c) Total orders



(d) Total actual revenue

Figure 1: Stylized Facts: Treatment Year (2025) versus Control Year (2024)

Notes. Each panel plots a daily series over the March–September window. Gray denotes the control year (2024) and red the treatment year (2025); the vertical dashed line marks the April 30, 2025 escalation of the subsidy war. The series coincide before April 30 and diverge afterward.

3.2 Empirical Strategy

3.2.1 Research Design

Our objective is to recover the causal effect of the 2025 delivery subsidy war on the operating performance of incumbent restaurants. The central inferential obstacle is that the subsidy war was a nationwide shock: from April 30, 2025 onward, essentially every merchant in our data was exposed to the platforms' aggressive pricing at the same time. There is therefore no contemporaneous set of untreated merchants from which to build a counterfactual. A comparison across merchants *within* 2025 is no remedy, because a restaurant's exposure to subsidies is itself an equilibrium outcome (merchants choose how aggressively to participate), so that any cross-sectional contrast would confound the treatment with selection.

We instead exploit the calendar structure of the shock. The war erupted on a sharply defined date, April 30, that was set by competition among the platforms rather than by conditions at any individual restaurant; and we observe the *same* merchants over the identical March–September window of the preceding, comparatively stable year, 2024. This yields a difference-in-differences (DiD) design in which the treatment is "operating during the 2025 war" and the counterfactual for a merchant's post-April-30 outcomes in 2025 is assembled from (i) that merchant's own pre-April-30 outcomes in 2025 and (ii) the pre-to-post seasonal change observed for the same merchant over the same calendar window in 2024.

Formally, index merchants by i, located in city $c(i)$; let d denote the day of the year, aligned to a common calendar across years, and let $y \in \{2024, 2025\}$ denote the year. Define the treatment-year indicator $\text{Treat}_y = \mathbf{1}\{y = 2025\}$ and the post indicator $\text{Post}_d = \mathbf{1}\{d \geq \text{Apr } 30\}$; the DiD regressor is their interaction $\text{Post}_d \times \text{Treat}_y$. The parameter of interest is the coefficient on $\text{Post}_d \times \text{Treat}_y$, which contrasts the post-versus-pre change in 2025 with the post-versus-pre change in 2024 for the same merchants.

The estimand is the average treatment effect on the treated (ATT): the average effect of the subsidy war on the post-April-30 operating outcomes of merchants that operate throughout both sample years, relative to the seasonal path those outcomes would have followed absent the war. Two features of this estimand deserve emphasis. First, because treatment is universal in 2025, the ATT is a *general-equilibrium* object. It captures not only each merchant's direct response to subsidized delivery but also any reallocation of demand across merchants, for instance, the siphoning of customers toward subsidy-participating restaurants. This is precisely the policy-relevant magnitude, namely the net effect of the war on the incumbent restaurant sector, and it is what the cross-year design (unlike a within-year cross-merchant comparison) is able to identify. Second, because we require merchants to operate in both years (Section 3.1), the ATT is defined for surviving incumbents; the war's effect on entry and exit is a distinct margin that we examine separately in Section 3.3.

3.2.2 Specification

Our baseline specification models the conditional mean of merchant outcomes multiplicatively:

$$\mathbb{E}\left[\, Y_{idy} \mid \text{Post}_d \times \text{Treat}_y,\ H_{dy},\ \boldsymbol{\mu} \,\right] = \exp\Big(\beta\,(\text{Post}_d \times \text{Treat}_y) + H_{dy}'\boldsymbol{\lambda} + \delta_{id} + \gamma_{w(d)} + \theta_{c(i)y}\Big), \tag{1}$$

where Y_{idy} is an outcome for merchant i on day-of-year d in year y; β is the parameter of interest; δ_{id} are merchant-by-day-of-year fixed effects; $\gamma_{w(d)}$ are day-of-week fixed effects, with $w(d)$ the weekday of date d; $\theta_{c(i)y}$ are city-by-year fixed effects; and H_{dy} is a vector of holiday indicators described below.

The fixed effects operationalize the design and absorb the leading threats to cross-year comparability (Section 3.2.3). The merchant-by-day-of-year effects δ_{id} subsume time-invariant merchant characteristics (cuisine, location, ownership) *and* each merchant's idiosyncratic within-year seasonality (a hot-pot restaurant's winter peak, a campus eatery's term-time cycle), so that identification comes solely from the differential 2025-versus-2024 deviation on each calendar date. The city-by-year effects $\theta_{c(i)y}$ absorb city-level annual shocks (local macroeconomic conditions, weather, related policies) common to all merchants in a city. Their inclusion also absorbs the main effect of Treat_y, while δ_{id} absorbs the main effect of Post_d; hence β is identified purely from the interaction $\text{Post}_d \times \text{Treat}_y$ and requires no additional time-varying controls.

We estimate (1) by Poisson pseudo-maximum likelihood (PPML) for order counts and revenues. PPML is the natural estimator for these outcomes for three reasons. First, the outcomes are nonnegative and contain a non-trivial mass of zeros, days on which a merchant records no delivery (or no dine-in) sales, which a log transformation cannot accommodate without ad hoc adjustments. Second, the multiplicative structure in (1) yields coefficients that are directly interpretable as proportional effects: the war changes the expected outcome by $100 \times (e^{\beta} - 1)$ percent, or approximately 100β percent for small β. Third, PPML is consistent and computationally feasible under the high-dimensional fixed effects we require (Santos Silva and Tenreyro, 2006; Correia et al., 2020). For outcomes that are intrinsically ratios or are defined only on the intensive margin (log unit prices and the dine-in shares of orders and revenue), we estimate by ordinary least squares the linear analogue of (1), with the identical fixed-effects structure:

$$\tilde{Y}_{idy} = \beta\,(\text{Post}_d \times \text{Treat}_y) + H_{dy}'\boldsymbol{\lambda} + \delta_{id} + \gamma_{w(d)} + \theta_{c(i)y} + \varepsilon_{idy}, \tag{2}$$

where $\tilde{Y}_{idy}$ denotes a log unit price or a dine-in share, the fixed effects and holiday controls are exactly those of (1), and ε_{idy} is an idiosyncratic error. For these outcomes β measures the war's effect in log points (for prices) or in share points (for the dine-in shares), rather than the proportional effect delivered by (1).

Throughout, we cluster standard errors at the city-by-calendar-month level, the level at which platform competitive intensity and local demand shocks plausibly co-move (Cameron and Miller, 2015). This choice yields a large number of clusters, so that cluster-robust inference is well behaved.

3.2.3 Identifying Assumptions and Threats to Validity

Identification rests on a conditional parallel-trends assumption: absent the subsidy war, the post-April-30 change in merchant outcomes in 2025 would, on average, have replicated the change observed over the same calendar window in 2024, after conditioning on the fixed effects and controls in (1). We first offer prima facie support for this assumption and then discuss the leading threats to it and how the design addresses each.

A first, informal look supports the assumption. Figure 1 plots the 2025 and 2024 daily series side by side for four key outcomes; in every panel the two years track each other closely over the pre-war window and separate only after April 30. This coincidence is prima facie evidence: a descriptive counterpart to parallel trends, not a substitute for a formal pre-trend test. With millions of merchant-days even small pre-war gaps would escape the eye, so residual confounders cannot be ruled out from the figure alone. The most consequential such confounder is a contemporaneous, economy-wide shock to restaurant demand in 2025 common to all merchants, which the cross-year design cannot by itself separate from the war. The triple-difference design of Section 3.4 partially addresses this concern for the channel-substitution outcomes, differencing out any 2025 shock common to delivery-capable and dine-in-only merchants.

Weekly seasonality and the leap-year misalignment. Restaurant demand follows a pronounced weekly cycle, with weekend peaks. Because 2024 is a leap year, aligning the two years on the calendar date breaks the correspondence between calendar date and day of the week: a given calendar date falls on a different weekday in 2024 than in 2025. Left unaddressed, this misalignment would mechanically contaminate the year-over-year comparison, since the treatment and control observations for a calendar date would mix different weekday compositions. We therefore include day-of-week fixed effects $\gamma_{w(d)}$, which absorb the weekly cycle and net out the leap-year shift so that β is not driven by calendar artifacts.

Movable holidays. Several major Chinese holidays follow the lunar calendar and thus fall on different Gregorian dates across years. In our window, the Dragon Boat and Mid-Autumn festivals occupy different positions in 2024 than in 2025, creating spikes in activity that do not line up across the two years. We control for these with a vector of holiday indicators H_{dy} that flag the affected days in each year, preventing holiday timing from loading onto the treatment coefficient.

Anticipation. The design assumes no anticipatory response before April 30. Individual restaurants are price-takers in the platform pricing game, and the escalation date was determined by competition among the platforms rather than announced to or negotiated with merchants; anticipatory behavior at the merchant level is therefore unlikely to bias the estimates.

Sample selection. Because the balanced panel conditions on operating in both years, the ATT pertains to surviving incumbents and is silent, by construction, on merchants that exit. Rather than treat this as a nuisance, we study the extensive margin (active days and permanent exit) as a separate outcome in Section 3.3, so that the survival and intensive-use responses to the war are reported alongside one another.

3.3 Main Empirical Results

This subsection presents the baseline estimates of equation (1), following the economics of the shock. We first document the direct effect of the subsidy war on the delivery channel, where the platforms cut prices (Section 3.3.1). We then show that the shock spilled over to the offline dine-in channel (Section 3.3.2). Aggregating the two channels yields the central finding, a divergence between order volume and revenue that we term the traffic–revenue paradox (Section 3.3.3). We then decompose the aggregate effect into extensive and intensive margins (Section 3.3.4); Section 3.4 isolates the channel-substitution mechanism, and Appendix A documents heterogeneity across merchants. Throughout, coefficients on order counts and revenues are estimated by PPML and read as proportional effects of approximately $100\widehat{\beta}$ percent; coefficients on log prices and on channel shares are OLS.[6]

3.3.1 Effects on the Delivery Channel

Table 2 reports the effect of the subsidy war on the delivery channel. The consumer-facing prediction is confirmed sharply. The war lowers the actual (post-subsidy) delivery price by 15.0 log points (column 1) and raises delivery order volume by 15.9 percent (column 2), both significant at the one percent level. Lower prices indeed buy more orders.

The volume gain does not, however, translate into delivery revenue. Column (3) shows that actual delivery revenue *falls* by 5.6 percent: the price cut more than offsets the order increase. The

[6]For PPML the exact proportional effect is $100\,(e^{\widehat{\beta}} - 1)$ percent, which coincides with $100\widehat{\beta}$ to first order and is marginally larger in absolute value for the larger coefficients. Share outcomes are levels, so their coefficients are in percentage points.

Table 2: The Delivery Channel

Dependent variable	(1) Log delivery price	(2) Delivery orders	(3) Delivery actual revenue
Post × Treat	−0.150***	0.159***	−0.056***
	(0.003)	(0.010)	(0.007)
Observations	18,670,638	19,310,966	19,310,966
Adj./Pseudo R^2	0.880	0.811	0.897
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
Estimator	OLS	PPML	PPML

Notes. Estimates of equation (1). Fixed effects: merchant × day-of-year, day-of-week, and city × year. Controls: movable-holiday indicators. Standard errors clustered at the city × calendar-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

"price-for-volume" bargain that subsidies impose on the delivery channel thus fails to enlarge, and in fact shrinks, the revenue merchants collect from it. This is the first piece of evidence that the order boom is not a revenue boom.

3.3.2 Effects on the Dine-in Channel

A maintained presumption in the two-sided-market benchmark is that merchants are insulated from platform pricing by their offline outside option: customers who do not order delivery can still dine in.[7] Table 3 shows that, far from being insulated, the dine-in channel contracts. Dine-in orders fall by 11.9 percent (column 1) and dine-in actual revenue by 7.5 percent (column 2). The composition of business shifts toward delivery on both counts: the dine-in share of orders falls by 4.6 percentage points (column 3) and the dine-in share of revenue by 1.6 percentage points (column 4).

These magnitudes are economically large: the contraction in dine-in orders is comparable in absolute value to the expansion in delivery orders. The pattern is the signature of *channel substitution*, subsidized delivery draws customers away from the dine-in counter, rather than of pure market expansion. We isolate this mechanism directly in Section 3.4.

[7] This is the competitive-bottleneck view of platform competition, in which platforms compete to attract consumers and recover rents from the merchant side, while merchants are shielded by an offline outside option (Rochet and Tirole, 2003; Armstrong, 2006; Caillaud and Jullien, 2003). Our evidence shows that when the price war is waged on the very channel that substitutes for that outside option, the protection it is presumed to provide breaks down.

Table 3: The Dine-in Channel

Dependent variable	(1) Dine-in orders	(2) Dine-in actual revenue	(3) Dine-in order share	(4) Dine-in revenue share
Post × Treat	-0.119^{***}	-0.075^{***}	-0.046^{***}	-0.016^{***}
	(0.004)	(0.005)	(0.002)	(0.001)
Observations	19,310,966	19,310,966	18,670,638	18,670,638
Adj./Pseudo R^2	0.852	0.951	0.783	0.817
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Estimator	PPML	PPML	OLS	OLS

Notes. See Table 2. Share outcomes are in levels; their coefficients are in percentage points. Standard errors clustered at the city × calendar-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.3.3 The Net Effect

Aggregating the two channels delivers the paper's central result, reported in Table 4. Total order volume rises by a modest 1.8 percent (column 1), confirming that the war does generate incremental traffic. Yet total *actual* revenue falls by 6.7 percent (column 2). Merchants handle more orders but take home substantially less money: the traffic–revenue paradox.

Column (3) clarifies who bears the cost. Total *listed* revenue, what customers would owe at posted menu prices, declines by only 1.5 percent, far less than the 6.7 percent drop in actual revenue. The roughly five-percentage-point wedge between listed and actual revenue is the portion of the price cut absorbed by merchants rather than reflected in nominal prices: the incidence of the subsidies falls on the merchant side of the market. The paradox is therefore not a mechanical artifact of discounting but a transfer of surplus away from merchants, operating through the channel-substitution mechanism documented next.

3.3.4 Extensive versus Intensive Margins

The aggregate revenue decline could reflect fewer merchants operating, lower activity per operating merchant, or both. Table 5 decomposes it. Panel A reports a linear probability model for whether a merchant is active on a given day. The probability of activity falls by 5.3 percentage points; consistent with this, permanent exit is non-negligible but small: 0.73 percent of merchants exit on a ≥ 14-day-gap definition and 0.04 percent on a stricter ≥ 30-day definition.

Panel B conditions on active days and re-estimates the log outcomes by OLS. Among continuously operating merchants, total orders per active day rise by 5.6 percent while actual revenue per active day still falls by 2.6 percent. The order increase is thus an intensive-margin phenomenon,

Table 4: Net Effect: The Traffic–Revenue Paradox

Dependent variable	(1) Total orders	(2) Total actual revenue	(3) Total listed revenue
Post × Treat	0.018***	−0.067***	−0.015***
	(0.005)	(0.004)	(0.004)
Observations	19,310,966	19,310,966	19,310,966
Pseudo R^2	0.836	0.942	0.935
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
Estimator	PPML	PPML	PPML

Notes. See Table 2. "Listed" revenue values orders at posted menu prices; "actual" revenue is post-subsidy receipts. Standard errors clustered at the city × calendar-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

more orders on days a merchant is open, whereas revenue contracts on *both* margins: merchants are active on fewer days, and earn less on the days they are active. The traffic–revenue paradox survives the decomposition.

3.4 Isolating Channel Substitution: Delivery-Capable versus Dine-in-Only Merchants

The baseline results above are consistent with a single mechanism: the subsidy war diverts demand from the high-margin dine-in counter to the low-margin delivery channel *within* the same merchant, so that order volume rises even as revenue falls. This section provides direct evidence for that interpretation and, in doing so, addresses a residual identification concern with the cross-year baseline. The challenge is that the baseline ATT confounds two forces that move dine-in business in the same direction, genuine channel substitution and any contemporaneous macroeconomic softening of restaurant demand in 2025. We separate them by exploiting a structural feature of the merchant population: a subset of restaurants operates no delivery channel at all and is therefore insulated, by construction, from the *within-merchant* substitution margin.

3.4.1 A Triple-Difference Design

Let $\text{HasDelivery}_i = \mathbf{1}\{\text{merchant } i \text{ operates a delivery channel}\}$, a structural attribute of the merchant rather than an endogenous response to the war.[8] We augment the baseline with the full set

[8] HasDelivery_i is the complement of a dine-in-only indicator. Because it reflects whether the merchant has a delivery operation at all, a feature of its business model, it is not a "bad control" in the sense of being chosen in response to the

Table 5: Extensive versus Intensive Margins

	Total orders	Total actual revenue
Panel A. Extensive margin: any active day (LPM)		
Post × Treat	-0.053^{***}	-0.053^{***}
	(0.002)	(0.002)
Observations	19,328,480	19,328,480
Adjusted R^2	0.068	0.068
Panel B. Intensive margin: log outcomes on active days (OLS)		
Post × Treat	0.056^{***}	-0.026^{***}
	(0.003)	(0.002)
Observations	18,670,638	18,670,638
Adjusted R^2	0.851	0.898

Notes. Panel A is a linear probability model for an indicator that the merchant records positive activity on a given merchant-day. Panel B estimates log outcomes on the sample of active merchant-days. Fixed effects and controls as in Table 2; standard errors clustered at the city × calendar-month level in parentheses. Permanent exit (not tabulated) affects 0.73% of merchants under a ≥ 14-day-gap definition and 0.04% under a ≥ 30-day definition. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

of interactions between HasDelivery_i and the DiD building blocks and estimate it by PPML on the order-count and revenue outcomes, retaining the fixed-effects structure of equation (1):

$$\mathbb{E}\left[\, Y_{idy} \mid \cdot \,\right] = \exp\Big(\beta_3 \left(\text{HasDelivery}_i \times \text{Post}_d \times \text{Treat}_y\right) + \beta_1 \left(\text{Post}_d \times \text{Treat}_y\right) + \zeta \left(\text{HasDelivery}_i \times \text{Post}_d\right) + \eta \left(\text{HasDelivery}_i \times \text{Treat}_y\right) + H'_{dy}\boldsymbol{\lambda} + \delta_{id} + \gamma_{w(d)} + \theta_{c(i)y}\Big). \tag{3}$$

The coefficient β_1 on the simple DiD term $\text{Post}_d \times \text{Treat}_y$ is the war's effect on dine-in-only merchants, who cannot substitute toward their own delivery channel. The triple-difference coefficient β_3 is the *additional* effect borne by delivery-capable merchants. It is identified from a comparison of comparisons (the 2025-versus-2024 change for delivery-capable merchants relative to the same change for dine-in-only merchants) and therefore differences out any 2025 shock common to all restaurants in a city. The dine-in-only group thus supplies the contemporaneous, within-year control that the cross-year baseline of Section 3.2 necessarily lacks, and β_3 is robust to violations of cross-year parallel trends that are common across the two merchant types.

Two caveats fix what this comparison can and cannot deliver. First, the dine-in-only control is not strictly untreated: if subsidized delivery at competing restaurants siphons walk-in diners, dine-in-only merchants also lose business, so β_3 understates the absolute within-merchant substitution

subsidies, unlike the intensity of subsidy participation we study in Section A.

and is best read as a lower bound. Second, identification of β_3 maintains that the two merchant types would have followed parallel dine-in paths absent the war. The differencing-out property neutralizes any 2025 shock common to both types but not one that hits them differentially; we verify these differential pre-trends directly with a triple-difference event study in ongoing work, and treat the present estimates as corroborating evidence under that maintained assumption.

3.4.2 Results

Table 6 reports estimates of equation (3) for total orders, total actual revenue, dine-in orders, and dine-in actual revenue.

The substitution mechanism is visible most directly in the dine-in outcomes (columns 3 and 4). The triple-difference coefficient is large and negative: relative to dine-in-only merchants, delivery-capable merchants lose an additional 9.4 percent of dine-in orders and 6.1 percent of dine-in revenue, both significant at the one percent level. The dine-in contraction is thus overwhelmingly concentrated among merchants that actually operate a delivery channel, precisely the prediction of within-merchant substitution. Had the dine-in decline instead reflected a uniform downturn in restaurant demand, it would have fallen equally on both merchant types and β_3 would be zero.

The mirror image appears in total orders (column 1). Delivery-capable merchants enjoy a traffic gain of 3.9 percent relative to dine-in-only merchants: the subsidized delivery channel is where the incremental orders accrue. Yet this traffic gain coincides with a *larger* revenue loss. In column (2), delivery-capable merchants lose an additional 5.1 percent of total revenue. The traffic–revenue paradox is therefore borne by exactly the merchants exposed to the substitution margin: those who gain the most orders lose the most money.

The remaining rows play two roles. The implied war effect for delivery-capable merchants, the sum $\beta_1 + \beta_3$, tracks the baseline ATT of Section 3.3: $-0.021 + 0.039 = +0.018$ for total orders, $-0.012 - 0.051 = -0.063$ for total revenue, and $-0.023 - 0.094 = -0.117$ for dine-in orders, against baseline estimates of +0.018, −0.067, and −0.119. Because delivery-capable merchants are the same population as the baseline sample, this agreement is largely mechanical; we read it as a consistency check that adding the dine-in-only arm leaves the baseline intact, not as independent corroboration. The substantive feature is instead the simple-DiD coefficient β_1 on dine-in-only merchants, itself small but statistically significant and negative across all outcomes (e.g. −0.023 for dine-in orders). Even restaurants that operate no delivery channel see their business contract modestly during the war. This residual decline is consistent with a general-equilibrium siphoning effect: subsidized delivery at competing restaurants draws customers away from dine-in-only incumbents, eroding the outside option even for non-participants, a theme echoed in the heterogeneity patterns below.

Table 6: Channel Substitution: Triple-Difference Estimates

Dependent variable	(1) Total orders	(2) Total actual revenue	(3) Dine-in orders	(4) Dine-in actual revenue
HasDelivery × Post × Treat	0.039***	−0.051***	−0.094***	−0.061***
	(0.005)	(0.004)	(0.004)	(0.005)
Post × Treat	−0.021***	−0.012***	−0.023***	−0.012***
	(0.003)	(0.002)	(0.002)	(0.002)
HasDelivery × Post	0.435***	0.171	0.411***	0.091
	(0.059)	(0.116)	(0.073)	(0.131)
HasDelivery × Treat	0.112***	0.083***	0.056***	0.053***
	(0.004)	(0.004)	(0.003)	(0.004)
Observations	78,667,739	78,667,739	78,667,739	78,667,739
Pseudo R^2	0.898	0.934	0.880	0.936
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes

Notes. PPML estimates of equation (3). HasDelivery_i indicates that merchant i operates a delivery channel; the omitted category is dine-in-only merchants. The sample pools delivery-capable and dine-in-only merchants and is therefore broader than the baseline of Table 4. Fixed effects: merchant × day-of-year, day-of-week, and city × year. Controls: movable-holiday indicators. Standard errors clustered at the city × calendar-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Summary. Taken together, the empirical evidence paints a clear picture of the subsidy war's distributional consequences. Consumers benefit from lower out-of-pocket prices; platforms suffer from unsustainable cash burn; and merchants lose on net. The merchant-level impact is particularly telling. While the subsidy war did what it was designed to do on the delivery channel—raising online order volume—it also depressed online prices. Crucially, the surge in delivery traffic did not represent new demand. Instead, it cannibalized high-margin offline dine-in orders, leading to a net decline in both total revenue and profits.

Yet, these empirical patterns, however clean, represent only the first step. The data alone cannot resolve the broader welfare calculus or guide policy. Specifically, three questions remain. First, does platform competition amplify or mitigate pre-existing pricing distortions? Second, what is the net effect on total surplus—is it positive, negative, or near zero? Third, is the regulatory policy of curbing consumer subsidies welfare-improving? To address these questions, we require a theoretical framework that models the strategic interactions among platforms, consumers, and merchants. Section 4 develops this model.

4 The Model

We now build a model that rationalizes the traffic–revenue paradox and lets us trace its welfare incidence. The model keeps the offline channel front and center: each merchant always has a storefront, and the value of that storefront is not a fixed outside option but an equilibrium object that the platforms' conduct moves. We present the setup (Section 4.1), define the two market structures that bracket the war (Section 4.2), and state the main results and comparative statics (Section 4.3). To keep the main text readable we keep the derivations compact and explain the economics as we go; the longer proofs are in the Online Appendix.

4.1 Basic Setup

Geography and agents. The economy is a continuum of identical islands, indexed by $i \in [0, N]$. Each island has a unit mass of consumers and a single merchant. Every merchant runs a brick-and-mortar storefront on its own island and may also list on an online platform. A platform governs online trade with two instruments: an ad-valorem commission $r \in [0, 1)$ charged to merchants, and a per-order transfer s paid to consumers, where $s > 0$ is a subsidy and $s < 0$ a fee.

Merchants. Merchants produce at a constant marginal cost c, the same online and offline. Merchant i sets an online price $p_{i,\mathrm{on}}$ and, separately, an offline price $p_{i,\mathrm{off}}$. To list online, it must pay a fixed participation cost φ_i—the overhead of an online operation, such as menu setup, payment integration, and order handling. This cost is drawn from a distribution G with density g on $[\underline{\varphi}, \overline{\varphi}]$.

Assumption 1 (Regular cost distribution)**.** G is log-concave, and the participation elasticity $\varepsilon(\varphi) \equiv \varphi\, g(\varphi)/G(\varphi)$ is strictly decreasing in φ.

Log-concavity is a standard property of common distributions and makes the virtual cost $\varphi + G(\varphi)/g(\varphi)$ increasing. Decreasing elasticity captures a natural idea: as the platform raises its participation threshold and exhausts the low-cost merchants, ever fewer new merchants sit near the margin.

Consumers. Consumer choice follows a two-level nested logit. At the lower level, a consumer who shops online chooses among the participating online merchants. Buying from online merchant i yields utility

$$u_i = v - p_{i,\text{on}} + s + \epsilon_i, \qquad \epsilon_i \sim \text{i.i.d. Type I EV}(0, \mu), \qquad \mu > 0, \tag{4}$$

where μ is the taste dispersion within the online nest. The expected maximum utility from shopping online is the inclusive value $I_{\text{on}} = \mu \ln \sum_i e^{(v - p_{i,\text{on}} + s)/\mu}$. At the upper level, the consumer chooses between the online channel and the offline storefront. The channel-level utilities are $U_k = I_k + \eta_k$ for $k \in \{\text{on}, \text{off}\}$, with channel shocks $\eta_k \sim \text{Type I EV}(0, \sigma)$ and the offline inclusive value $I_{\text{off}} = v - p_{i,\text{off}}$. We impose the nesting condition

$$\mu \leq \sigma, \tag{5}$$

so that two online merchants are closer substitutes for each other than the online channel is for the storefront.

Timing. The game has three stages. First, the platform(s) set the policy (s, r). Second, merchants draw their participation costs, decide whether to list, and set online and offline prices. Third, consumers observe prices and choose a channel and a merchant. We study symmetric subgame-perfect equilibria.

Demand. We derive demand here for the simplest case—a single platform alongside the offline channel—and carry the resulting objects into both duopoly structures below, where the only change is the upper nest (a binary nest per captive segment, or a single ternary nest). Let $\mathcal{M}$ be the set of listing merchants, of measure $n = N\, G(\hat{\varphi})$ under the threshold rule derived shortly. A consumer's online inclusive value is $I_{\text{on}} = \mu \ln \int_{\mathcal{M}} e^{(v - p_{j,\text{on}} + s)/\mu}\, dj$, and her offline value on island

i is $I^{(i)}_{\text{off}} = v - p_{i,\text{off}}$. The share of island-$i$ consumers who shop online is the upper-level logit

$$S^{(i)} = \frac{e^{I_{\text{on}}/\sigma}}{e^{I_{\text{on}}/\sigma} + e^{I^{(i)}_{\text{off}}/\sigma}}, \qquad S^{(i)}_0 = 1 - S^{(i)}, \tag{6}$$

and S denotes the average of $S^{(i)}$ across islands. A participating merchant's online demand is $D_i = NS\, f(i \mid \text{on})$, where $f(i \mid \text{on})$ is the lower-nest conditional choice density, and its offline demand is $D_{i,\text{off}} = S^{(i)}_0$. Because each merchant is atomistic, its own prices leave every aggregate (I_{on}, S, n) unchanged; only its own conditional density and its own offline share respond.

In a symmetric equilibrium all listing merchants set a common p_{on} and p_{off}, so the inclusive values collapse to

$$I_{\text{on}} = v - p_{\text{on}} + s + \mu \ln n, \qquad I_{\text{off}} = v - p_{\text{off}}, \tag{7}$$

and the upper logit gives the log-odds relation $S/(1-S) = e^{(I_{\text{on}} - I_{\text{off}})/\sigma}$. Inverting it expresses the subsidy required to implement a target online share S,

$$s = \sigma \ln \frac{S}{1-S} + p_{\text{on}} - p_{\text{off}} - \mu \ln n. \tag{8}$$

We abstract from the no-purchase option, so the market is fully covered. With two platforms the same logic applies: under the *captive* structure each platform faces a binary nest $\{k, \text{off}\}$ over its own consumer segment, and (8) holds segment by segment; under the *contestable* structure all consumers share one ternary nest $\{1, 2, \text{off}\}$, and the offline odds are taken against the residual offline share. We turn to these two structures in Section 4.2.

Merchant pricing: two markups. The merchant's pricing problem is the same in every market structure, so we state it once. Because each merchant is atomistic, its own prices do not move any aggregate. Maximizing online profit $[(1-r)p_{i,\text{on}} - c]\, D_i$ gives the online markup and price

$$m_{\text{on}} = \mu(1-r), \qquad p_{\text{on}} = \mu + \frac{c}{1-r}. \tag{9}$$

The online markup is the logit brand rent μ, thinned by the commission. Maximizing offline profit $(p_{i,\text{off}} - c)\, S^{(i)}_0$ gives the offline markup and price

$$m_{\text{off}} = \frac{\sigma}{T}, \qquad p_{\text{off}} = c + \frac{\sigma}{T}. \tag{10}$$

Here T is total online penetration. Its explicit form differs across the structures below—under monopoly $T = S$, in the captive duopoly $T = \frac{1}{2}(S_1 + S_2)$, and in the contestable duopoly $T = S_1 + S_2$—but it always denotes the same object: the fraction of the entire consumer population that transacts online on one of the platforms. The economics of (10) is as follows. The storefront is a local monopoly, but it competes against the online channel for every consumer. As penetration T rises, the marginal offline customer is easier to lure online, offline demand grows more elastic, and the storefront's optimal markup σ/T falls. *The outside option is endogenous*: its value is set by how far the platforms have pushed consumers online. When channels are poor substitutes (σ large) the storefront keeps a fat markup; the platform cannot easily erode the merchant's fallback.

Participation. Consider merchant i's decision to list. Because it is atomistic, listing does not change its offline profit; it joins if and only if the online business covers the fixed cost. With online volume per listing merchant equal to NS/n, the condition is

$$m_{\text{on}} \frac{NS}{n} \geq \varphi_i. \tag{11}$$

The marginal merchant has cost $\hat{\varphi}$ at which (11) binds, $m_{\text{on}} NS/n = \hat{\varphi}$. Substituting $n = N\,G(\hat{\varphi})$ gives the participation cutoff

$$\hat{\varphi}\,G(\hat{\varphi}) = m_{\text{on}}\,S, \qquad n = N\,G(\hat{\varphi}). \tag{12}$$

Merchants with $\varphi_i \leq \hat{\varphi}$ list; the rest stay offline only. The platform sets (s, r), but it is more convenient to let it choose the real allocation $(S, \hat{\varphi})$ directly; this change of variables, and the equilibrium commission and subsidy it delivers, are derived inside each market structure below.

4.2 Two Market Structures: Captive and Contestable

The war changed how consumers shop, not what they like. Before the war, most consumers used a single app and never compared platforms. During the war, subsidies and aggressive promotion made comparison routine. We capture this with two market structures that share all primitives and differ only in whether consumers compare platforms. The pre-war regime is the *captive* duopoly; the war regime is the *contestable* duopoly.

4.2.1 Captive Duopoly

Two symmetric platforms, $k \in \{1, 2\}$, each set a policy (s_k, r_k). On every island the unit consumer mass splits into two equal, disjoint, fixed segments: a segment-1 consumer chooses only between platform 1 and the storefront, a segment-2 consumer only between platform 2 and the storefront. Each platform is a local monopolist over its captive half, of aggregate mass $N/2$. The platforms never poach each other directly; they are linked only through the single storefront, which serves both segments. Because each platform reaches only half of each island, a merchant that wants the whole island must list on both—it multi-homes—and pays the fixed cost twice. The per-platform listing benefit is therefore halved, and the participation cutoff (12) becomes

$$\hat{\varphi}_k \, G(\hat{\varphi}_k) = \tfrac{1}{2}\, m_{\text{on}}\, S_k. \tag{13}$$

Relative to a single platform, fragmentation duplicates the listing cost and shrinks the active merchant base.

From prices to allocations. Platform k earns $\Pi_k = (r_k p_k - s_k)\frac{N}{2} S_k$. We show that the map from its instruments (s_k, r_k) to the real allocation $(S_k, \hat{\varphi}_k)$ is a bijection. The platform may therefore choose $(S_k, \hat{\varphi}_k)$ directly. The per-segment profit $\pi_k \equiv (r_k p_k - s_k) S_k$ can be written as

$$\pi_k(S_k, S_{k'}, \hat{\varphi}_k) = \underbrace{\frac{2\sigma(1-T)\, S_k}{S_k(1-S_k) + S_{k'}(1-S_{k'})}}_{\text{offline rent}} + \mu S_k \ln n_k - \sigma S_k \ln \frac{S_k}{1-S_k} - 2\hat{\varphi}_k \, G(\hat{\varphi}_k), \tag{14}$$

where $T = \frac{1}{2}(S_1 + S_2)$.

Merchant admission and the commission. The first-order condition $\partial \pi_k / \partial \hat{\varphi}_k = 0$ is

$$\mu S_k = 2\, G(\hat{\varphi}_k)\left[\hat{\varphi}_k + \frac{G(\hat{\varphi}_k)}{g(\hat{\varphi}_k)}\right]. \tag{15}$$

Combining (15) with the binding cutoff (13) and $m_{\text{on}} = \mu(1-r)$ delivers the equilibrium commission as an inverse-elasticity rule,

$$1 - r_c^* = \frac{\varepsilon_c^*}{1+\varepsilon_c^*}, \qquad \text{equivalently} \quad r_c^* = \frac{1}{1+\varepsilon_c^*}, \qquad \varepsilon_c^* \equiv \frac{\hat{\varphi}_c^* g(\hat{\varphi}_c^*)}{G(\hat{\varphi}_c^*)}. \tag{16}$$

The platform and the merchants split the brand rent μ in the ratio $r/(1-r) = 1/\varepsilon$, a monopsony Lerner rule against the merchant supply curve. We will see that this rule keeps the same *form* in the contestable regime; what changes across regimes is the elasticity ε^*, and hence the level of r^*.

The consumer margin: a cross-platform externality. When platform 1 expands S_1, total penetration T rises, the storefront cuts $p_{\text{off}} = c + \sigma/T$ to defend its base, and—because the storefront also serves segment 2—the cheaper dine-in option pulls some of platform 2's consumers offline. Holding (s_2, r_2) fixed, this gives a rival response along the deviation path

$$a_c \equiv \frac{dS_2}{dS_1} = -\frac{S\, Q^c_{\text{cross}}}{\theta + S\, Q^c_{\text{own}}} \in (-1, 0), \tag{17}$$

where $Q^c_{\text{cross}} = \frac{1}{2T^2}$, $Q^c_{\text{own}} = \frac{1}{1-T} + \frac{1}{2T^2}$, and $\theta = 1 - \frac{\mu}{\sigma}\frac{\varepsilon}{1+\varepsilon}$. The cross-shock $Q^c_{\text{cross}} = 1/(2T^2)$ travels purely through the shared storefront price, while the rival absorbs it through both its own logit base and the storefront. Differentiating the profit (14) in S_k along this path (with $\hat{\varphi}_k$ at its optimum) gives the share first-order condition

$$\frac{\mu \ln n}{\sigma} = h_c(S) \equiv \ln \frac{S}{1-S} + \frac{1}{1-S} - \frac{1-a_c}{2S}. \tag{18}$$

Once (15) and (18) pin down $(\hat{\varphi}_c, S_c)$, the instruments follow. The commission is (16), and the subsidy comes from the inversion evaluated at the equilibrium prices $p_c = \mu + c/(1-r_c)$ and $p_{\text{off}} = c + \sigma/S_c$:

$$s_c = \mu + \frac{c}{\varepsilon_c} - \frac{\sigma}{S_c} - \frac{\sigma}{1-S_c} + \frac{\sigma(1-a_c)}{2S_c}. \tag{19}$$

4.2.2 Contestable Duopoly

Now the captive walls come down. Every consumer considers both platforms and the storefront, so the upper nest is a single ternary logit $\{1, 2, \text{off}\}$ over the whole consumer mass. A consumer won by platform 1 is one platform 2 could have served—the business-stealing margin, absent under captivity, is now live. The island shares are $S_k = e^{I_k/\sigma}/(e^{I_1/\sigma} + e^{I_2/\sigma} + e^{I_{\text{off}}/\sigma})$ with $T = S_1 + S_2$ and $S_0 = 1 - T$, and the demand inversion becomes

$$s_k = \sigma \ln \frac{S_k}{S_0} + p_k - p_{\text{off}} - \mu \ln n_k, \tag{20}$$

the only change from (8) being that the offline odds are taken against the residual $S_0 = 1 - T$ left after *both* platforms. Merchant pricing and participation are unchanged: $p_{\text{off}} = c + \sigma/T$ now holds for all (S_1, S_2), not just at symmetry, because the single storefront faces one offline share, and the cutoff is $\hat{\varphi}_k G(\hat{\varphi}_k) = m_{\text{on},k} S_k$.

Profit in allocations. Platform 1's profit per unit consumer mass is $\Pi_1 = (r_1 p_1 - s_1) S_1$. Substitute $r_1 p_1 = p_1 - (m_{\text{on},1} + c)$ and the inversion (20) to get $r_1 p_1 - s_1 = -m_{\text{on},1} - c + p_{\text{off}} - \sigma \ln \frac{S_1}{S_0} + \mu \ln n_1$; then use $p_{\text{off}} - c = m_{\text{off}} = \sigma/T$, multiply by S_1, and apply the cutoff $m_{\text{on},1} S_1 = \hat{\varphi}_1 G(\hat{\varphi}_1)$. This yields

$$\Pi_1(S_1, \hat{\varphi}_1) = \frac{\sigma}{T} S_1 + \mu S_1 \ln n_1 - \sigma S_1 \ln \frac{S_1}{S_0} - \hat{\varphi}_1 \, G(\hat{\varphi}_1). \tag{21}$$

This is the direct analogue of the captive profit (14), with two differences that come from the single shared pool: the offline rent is the clean $\frac{\sigma}{T} S_1$ (the markup σ/T now holds off symmetry too), and the merchant bill is $\hat{\varphi}_1 G$, since the acquisition cost is taken against $S_0 = 1 - T$ rather than $1 - S_1$.

Merchant admission and the commission. Holding S_1 fixed, the extensive-margin first-order condition is, exactly as before,

$$\mu S_1 = G(\hat{\varphi}_1)\Big[\hat{\varphi}_1 + \frac{G(\hat{\varphi}_1)}{g(\hat{\varphi}_1)}\Big], \tag{22}$$

and the commission again follows the inverse-elasticity rule $1 - r_w^* = \varepsilon_w^*/(1+\varepsilon_w^*)$, with ε_w^* the war participation elasticity. The form of (16) is preserved; the level of r^* moves only because ε^* does.

A stronger externality. The deviation path now carries an extra channel. Holding (s_2, r_2) fixed,

$$a_t \equiv \frac{dS_2}{dS_1} = -\frac{S_2\,Q}{\theta + S_2\,Q}, \tag{23}$$

where $Q = \frac{1}{S_0} + \frac{1}{T^2}$, with the same $\theta = 1 - \frac{\mu}{\sigma}\frac{\varepsilon}{1+\varepsilon}$. Writing the captive and contestable paths in total penetration T (recall $S_c = T$ but $S_w = T/2$) lays the two side by side:

$$a_c = -\frac{\frac{1}{2T}}{\theta + \frac{T}{S_0} + \frac{1}{2T}}, \qquad a_t = -\frac{\frac{1}{2T} + \frac{T}{2S_0}}{\theta + \frac{T}{2S_0} + \frac{1}{2T}}. \tag{24}$$

Two forces make the contraction fiercer under contestability. First, the numerator gains the direct poaching term $\frac{T}{2S_0}$: a consumer won is now taken straight from the rival's online demand, not merely lured through a cheaper storefront. Second, the rival's own-logit buffer in the denominator is halved (from $\frac{T}{S_0}$ to $\frac{T}{2S_0}$), because a contestable platform commands only half the online base and so has less captive mass to absorb the blow. Both effects strictly raise the magnitude, $|a_t| > |a_c|$. The share first-order condition becomes

$$\frac{\mu \ln n}{\sigma} = h_t(T), \qquad h_t(T) \equiv \ln\frac{T}{1-T} + 1 - \ln 2 + \frac{T(1+a_t)}{2(1-T)} - \frac{1-a_t}{2T}, \tag{25}$$

which lies strictly below the captive schedule $h_c(T)$ by at least $\sigma \ln 2$ per unit, the pure value of splitting one online channel into two, plus the extra poaching relief. The subsidy is again recovered from the inversion at the equilibrium, $s_w = \sigma \ln \frac{S_w}{S_0} + p_{\text{on}} - p_{\text{off}} - \mu \ln n_w$.

The two structures bracket the war: the pre-war market is captive, the war market is contestable. We now ask what changes when the market moves from one to the other.

4.3 Main Results and Comparative Statics

We collect the results that the calibration and the policy analysis use. Each holds for every cost distribution G satisfying Assumption 1 and every admissible $\mu \le \sigma$; none relies on functional-form choices made in the calibration. Proofs are in the Online Appendix.

Proposition 1 (Penetration, base, and the offline markup)**.** *Moving from the captive to the contestable duopoly strictly raises total online penetration, $T_w^* > T_c^*$, for every admissible G. Consequently the merchant base expands, $n_w^* > n_c^*$, and the storefront markup falls, $\sigma/T_w^* < \sigma/T_c^*$, so the offline rent pool $\sigma(1-T)/T$ shrinks.*

What happens, in words: when consumers begin to compare platforms, more of them end up buying online, more merchants find it worth listing, and the dine-in counter loses its pricing power. Why it happens follows from the share conditions (18) and (25). Contestability adds the business-stealing margin, which lowers each platform's private cost of acquiring a consumer; formally the acquisition schedule drops from h_c to h_t, by at least $\sigma \ln 2$. With a lower marginal cost both platforms expand and penetration rises. Higher penetration feeds the merchant base through (12), and, through the conservation law (10), it drives the storefront markup down. This is the model's account of the dine-in erosion in the data: the merchant's outside option weakens precisely because the platforms succeed in pulling consumers online.

Proposition 2 (Prices and platform instruments)**.** *The pricing and commission* rules *are structural constants: in both regimes each merchant sets $p_{on} = \mu + c/(1-r)$ and $p_{off} = c + \sigma/T$, and the commission obeys $r^* = 1/(1+\varepsilon^*)$. Their* levels *move with the transition. From captive to contestable, both platform instruments rise—the consumer subsidy, $s_w^* > s_c^*$, and the commission, $r_w^* > r_c^*$—so the online menu price p_{on} rises while the storefront price $p_{off} = c + \sigma/T$ falls. The subsidy escalation is carried mainly by the erosion of the offline markup.*

The pricing rules are pinned by local, atomistic incentives—one merchant against the brand rent μ and against the storefront elasticity—which market structure does not touch. The levels move because the aggregates inside the rules do. In the inversion $s = \sigma \ln(S/S_0) + p_{\text{on}} - p_{\text{off}} - \mu \ln n$, the dominant moving part is $-p_{\text{off}} = -(c + \sigma/T)$: as penetration rises and the storefront markup collapses, the offline option becomes a weaker draw, and matching it takes a larger cash subsidy. The commission rises because, under Assumption 1, the larger merchant base lowers the participation elasticity ε^*, and the inverse-elasticity rule (16) then prescribes a higher take. Both contract terms a regulator can see thus move up, even as the real damage runs through a term—the offline markup—that no contract reports.

Proposition 3 (Who gains and who loses)**.** *In the move from the captive to the contestable duopoly:*

(i) combined platform profit strictly falls—the war is a prisoner's dilemma;

(ii) consumer surplus strictly rises;

(iii) merchant surplus falls, whenever the channel rent is large relative to the brand rent ($\sigma \ln 2 \geq \mu$, the empirically relevant case), because the collapse of the offline rent pool dominates the merchant's gain from a larger online business.

Part (i) is the self-inflicted wound. At any common penetration the two-platform configuration is actually *more* profitable than one, by the option value $\sigma \ln 2$ per online consumer. The platforms lose not because two channels are worse than one, but because the business-stealing externality drives them to over-penetrate past the point that maximizes their joint profit; each platform's first-order condition ignores the harm it does to the rival. Part (ii) follows from the conservation law: a lower storefront markup means cheaper offline purchases and a more valuable online option, both of which raise consumer surplus. Part (iii) is the merchant's predicament. Merchants do collect more online—the base is larger—but the online gain is bounded, while the offline rent pool $\sigma(1 - T)/T$ collapses as penetration rises. When the channel rent σ is large, the offline loss wins.

The three propositions add up to a single statement. A subsidy paid to consumers is, in its incidence, a tax on the merchant's outside option. Competition does not transfer surplus from platforms to merchants; it destroys the storefront markup that was the merchant's protection, and hands the proceeds to consumers. The mechanism is invisible on the terms regulators monitor: the pricing rules are fixed in form (Proposition 2), and the commission and the posted online price can even move the "wrong" way while merchant profit falls. The harm is real but off the books, because it lives in the offline markup, which competition quietly erodes. The next section puts numbers on these flows.

5 Quantitative Analysis

The model delivers sharp qualitative predictions, but the welfare question is quantitative: how large is the redistribution, and is the net efficiency effect positive or negative? To answer it we calibrate the model to the 2025 episode, treating the pre-war market as the captive duopoly and the war market as the contestable one. We map the data into structural parameters (Section 5.1), audit the welfare flows and compare them with the social optimum (Section 5.2), and simulate the policies regulators reached for (Section 5.3).

5.1 Calibration

Strategy. We hold the deep preference parameters—the brand rent μ, the channel rent σ, and the marginal cost c—fixed across the two regimes, so that any change in outcomes reflects the change in market structure rather than a change in tastes. We then proceed in three blocks. First, we read the preference parameters off the pre-war pricing identities. Second, we recover the merchant cost distribution from the participation margins on both sides of the war. Third, we solve the war regime as an equilibrium of the contestable game and let it predict the one number we have held back, the consumer subsidy. Two empirically necessary extensions enrich the bare model. First, the online and offline channels carry different intrinsic values, summarized by a premium $\Delta v \equiv v_{\text{on}} - v_{\text{off}}$, so that the storefront is a genuine alternative rather than a tie. Second, in the war consumers pay a private attention cost to compare platforms, which makes the contestable fraction itself an equilibrium outcome. We specify this second extension before turning to the data.

Comparison shopping and the search cost. The contestable duopoly of Section 4 assumes that every consumer compares both platforms. In practice, comparison takes effort—opening a second app and checking its prices—so we let it be a choice. Each consumer is assigned a default platform, with the two platforms splitting the defaults equally, and draws a private search cost $\psi \geq 0$ from a distribution $F(\psi;\kappa)$. A consumer who does not search chooses between her default platform and the storefront, a binary logit; a consumer who pays ψ learns both platforms' prices and then chooses among both platforms and the storefront, a ternary logit. Comparison pays whenever the search cost falls below the gain in expected match value from adding the rival platform to the choice set. Under symmetry that threshold is

$$\bar{\psi}(y) = \sigma \ln \frac{2y+1}{y+1}, \qquad y \equiv e^{(I_k - I_{\text{off}})/\sigma}, \tag{26}$$

where y is the online-versus-offline logit index. The share of consumers who search and compare, the *contestable fraction*, is then

$$\rho = F\big(\bar{\psi}(y)\big), \tag{27}$$

which interpolates between the captive market ($\rho = 0$, no one compares) and the fully contestable market ($\rho = 1$, everyone does). Averaging over default groups and search decisions, each platform's online share is

$$S = \rho \, \frac{y}{2y+1} + \frac{1-\rho}{2} \, \frac{y}{y+1}, \tag{28}$$

the first term collecting the ρ comparison shoppers (each choosing platform k in the ternary logit) and the second the $1-\rho$ loyal consumers (each split between the two default groups, choosing in a binary logit against the storefront). We take the search cost to be exponential, $F(\psi) = 1 - e^{-\psi/\kappa}$ with mean κ, a one-parameter family. This is the only primitive the bare model lacks, and κ is the single new parameter; we recover it in Step 4.

Data moments. Table 7 lists the targets. The two commissions are read directly from the gap between the online menu price and the merchant's receipt, $r = (p_{\text{on}} - w)/p_{\text{on}}$, giving $r_c = 0.200$

before the war and $r_w = 0.285$ during it. We set the marginal cost to $c = 24.0$ yuan, about a one-third ingredient share of the dine-in price. The single pre-war subsidy moment, $s_c = -3.50$, is used once, to identify the taste premium Δv; the war subsidy denoted by s_w is left for the model to predict.

Table 7: Observed Data Moments

Moment	Pre-war (captive)	War (contestable)
Online menu price per order p_{on}	47.78	46.00
Online receipt per order w	38.22	32.90
Dine-in price per order p_{off}	71.96	72.54
Total online share T	0.43	0.51
Observed consumer subsidy per order s	−3.50	-
Commission $r = (p_{\text{on}} - w)/p_{\text{on}}$	0.200	0.285
Participation elasticity $\varepsilon = (1 - r)/r$	4.00	2.511

Notes. All monetary objects are in yuan per unit consumer mass. Marginal cost is fixed exogenously at $c = 24.0$. The war subsidy is a model prediction (Step 4), not a calibration target.

Step 1: preference parameters. The pre-war pricing identities pin μ and σ directly. The online receipt obeys $w_c = \mu(1-r_c)+c$, so the brand rent is $\mu = (38.22-24.0)/0.80 = 17.78$. The dine-in price obeys $p_{\text{off},c} = c + \sigma/T_c$, so the channel rent is $\sigma = 0.43 \times (71.96 - 24.0) = 20.62$. Both are preference parameters, so we hold them fixed in the war.

Step 2: merchant cost distribution. We take G to be log-logistic with shape β and scale α. Each regime supplies two observables: the participation-cutoff value $A \equiv \hat{\varphi}\, G(\hat{\varphi})$ and the elasticity ε. The cutoff value is $A_c = \frac{1}{2} m_{\text{on},c} S_c = 3.057$ in the captive regime (the one-half reflects multi-homing) and $A_w = m_{\text{on},w} S_w = 3.242$ in the war. The two elasticities are $\varepsilon_c = 4.00$ and $\varepsilon_w = 2.511$. Requiring the *same* scale α in both regimes gives one equation that pins $\beta = 36.49$, and then $\alpha = 3.24$. The implied cutoffs and participation rates are $\hat{\varphi}_c = 3.43$, $\hat{\varphi}_w = 3.48$, $G(\hat{\varphi}_c) = 0.891$, and $G(\hat{\varphi}_w) = 0.932$. The total merchant pool follows from the pre-war base, $N = n_c/G(\hat{\varphi}_c) = 0.795$.

Step 3: the taste premium. The premium Δv is the residual that reconciles the pre-war channel split with observed prices. In the captive regime the index inversion gives $\Delta v = \sigma \ln y_c + p_{\text{on},c} - p_{\text{off},c} - s_c - \mu \ln n_c = -20.35$, where the captive merchant base $n_c = 0.708$ comes from the platform's first-order condition. The premium is large and negative: online is roughly 24 yuan cheaper on the menu, yet only 43 percent of consumers go online, which a logit at $\sigma \approx 20$ rationalizes only with a strong intrinsic preference for dining in. Like μ and σ, Δv is identified once before the war and held fixed.

Step 4: the war equilibrium, and a predicted subsidy. The war regime is calibrated entirely from the supply side. We require the observed penetration $T_w = 0.51$ (per-platform share $S_w = 0.255$) to be a symmetric static Nash equilibrium of the search-aware contestable game, and we let the model deliver the subsidy as a prediction rather than a target. Three of the war's objects are already fixed: the predicted online price $p_{\text{on},w} = \mu + c/(1 - r_w) = 51.33$, the predicted dine-in price $p_{\text{off},w} = c + \sigma/T_w = 64.43$, and the predicted merchant base $n_w = N\,G(\hat{\varphi}_w) = 0.740$. Three equations then pin the three remaining unknowns—the logit index y_w, the contestable fraction ρ_w, and the search-cost scale κ:

$$\text{(i) share accounting:} \quad S_w = \rho\,\frac{y}{2y+1} + \frac{1-\rho}{2}\,\frac{y}{y+1}; \tag{29}$$

$$\text{(ii) search fixed point:} \quad \rho = F\big(\bar{\psi}(y);\kappa\big) = 1 - e^{-\bar{\psi}(y)/\kappa}, \qquad \bar{\psi}(y) = \sigma \ln \frac{2y+1}{y+1}; \tag{30}$$

$$\text{(iii) platform optimality:} \quad \frac{\mu \ln n_w}{\sigma} = h_w(S_w;\rho,\kappa). \tag{31}$$

Equation (29) is the share mixture (28) evaluated at the observed penetration; (30) ties the contestable fraction to the same scale κ that enters the platform's problem; and (31) is the platform's intensive-margin first-order condition, whose right-hand side h_w is the war analogue of the consumer-acquisition cost of Section 4. The system is exactly identified. Solving it numerically yields

$$y_w = 0.723, \qquad \rho_w = 0.527, \qquad \kappa = 9.65 \text{ yuan}, \qquad \bar{\psi}(y_w) = 7.22 \text{ yuan}. \tag{32}$$

Two internal checks confirm the solution: the fixed point holds, $1 - e^{-7.22/9.65} = 0.527$, and the share adds up, $0.527 \times \frac{0.723}{2.446} + \frac{0.473}{2} \times \frac{0.723}{1.723} = 0.156 + 0.099 = 0.255$.

What ties the search distribution to the platform's optimum is a comparison-suppression, or *lock-in*, force in (31). When a platform raises its own subsidy it does two things: it attracts marginal consumers, and it discourages its own defaulters from searching, because a more attractive home platform lowers their incentive to open the rival's app. A consumer is worth more to the platform when she does *not* compare—her business is then less contestable—so this lock-in makes the platform's demand strictly more subsidy-elastic and raises the privately optimal subsidy. The force vanishes when the platform is search-myopic (formally $f \to 0$, a flat F), in which case the static subsidy collapses toward zero; it is precisely the curvature of F that sustains a positive subsidy. Reading the subsidy off the demand inversion (8) at the solution gives

$$s_w = \sigma \ln y_w - \Delta v + p_{\text{on},w} - p_{\text{off},w} - \mu \ln n_w = -6.69 + 20.35 + 51.33 - 64.43 + 5.35 = +5.92 \text{ yuan}, \tag{33}$$

a predicted subsidy of roughly six yuan per order. This closely matches the subsidy levels observed during the war. Because no subsidy data enter the calibration, this match is an out-of-sample validation of the lock-in mechanism, not a fitted moment. The prediction is robust: the implied subsidy stays in $[+4.9, +5.9]$ across $\kappa \in [6, 16]$ and peaks at the observed penetration $T_w = 0.51$.

5.2 Welfare Analysis

The flow audit. Table 8 reports surplus by party, evaluated at the model's predicted prices, under which the three pieces add exactly to total surplus. The pattern is stark. Consumers gain 9.30 yuan, merchants lose 7.38, and platforms lose 1.17. The war is, first and foremost, a redistribution from merchants and platforms to consumers, financed by the collapse of the offline markup. The net change in total surplus, +0.74, is small beside these transfers.

Table 8: Welfare Flow Audit

Party	Pre-war	War	Change
Consumers (CS)	39.63	48.93	+9.30
Merchants (Π^{mer})	28.91	21.53	−7.38
Platforms (Π^{plat})	5.61	4.44	−1.17
Total surplus (TS)	**74.15**	**74.89**	**+0.74**

Notes. Yuan per unit consumer mass, at the model's predicted prices and the FOC-predicted war subsidy $s_w = +5.92$. Total surplus is the transfer-free planner objective, so it is independent of how the subsidy is scored.

Where the merchant loss comes from. Decomposing merchant profit confirms the model's thesis. The loss is almost entirely the collapse of the offline rent pool, which falls from 27.34 to 19.81, a drop of 7.53. The online side is nearly flat: a slightly larger online business (+0.37) is offset by a marginally higher entry bill (−0.22). The storefront markup erodes (p_{off} falls from 71.96 to a predicted 64.43) and offline volume shrinks; that erosion, not the commission, is what hurts merchants. This is exactly the off-the-books harm of Proposition 3: a regulator watching commissions and posted prices would see almost nothing, while the merchant's profit falls by a quarter.

Comparison with the social optimum. To read the efficiency content we strip all transfers and compare each regime with its own social optimum. The social planner values the realized allocation at its real components only—channel match value, variety, and the option value of

Table 9: Decomposition of Merchant Profit

Term	Pre-war	War	Change
Offline rent $(p_{\text{off}} - c)(1 - T)$	27.34	19.81	−7.53
Online margin $(w - c)\,T$	6.11	6.48	+0.37
Entry cost	−4.55	−4.77	−0.22
Total	**28.91**	**21.53**	**−7.38**

comparison shopping, net of entry and search costs—with every markup, commission, and subsidy netted out as a transfer:

$$W = v + T\big(\Delta v + \mu \ln n\big) + \sigma \ln(y+1) + \int_0^{\bar{\psi}(y)} F(\psi)\,d\psi - T\sigma \ln y - c - 2N \int_0^{\hat{\varphi}} \varphi\, dG(\varphi). \quad (34)$$

Choosing the channel split and merchant entry to maximize (34), and using the envelope cancellations, yields two optimality conditions,

$$\underbrace{\sigma \ln y = \Delta v + \mu \ln n,}_{\text{channel}} \qquad \underbrace{\hat{\varphi}\, G(\hat{\varphi}) = \tfrac{1}{2}\mu T}_{\text{entry}}\,. \quad (35)$$

The channel condition prices the online option at its real worth—the taste premium Δv plus the variety $\mu \ln n$—with no markup and no subsidy. The entry condition admits merchants up to the point where the marginal participation cost equals the social variety benefit, without the monopsony wedge the platform imposes. Because the taste premium is large and negative ($\Delta v = -20.35$), the planner wants most consumers offline: the socially optimal penetration is only 7.3 percent before the war and 13.7 percent during it. The market, pulled online by subsidized prices against a marked-up storefront, instead reaches 43 and 51 percent. Both regimes thus suffer from online *over*-penetration, and both fall about two yuan short of their social optimum (Table 10). The war's distortion is the *smaller* of the two (−2.06 versus −2.42), because comparison shopping adds variety the planner credits to the online channel, raising the optimal target and shrinking the proportional over-shoot. The change in total surplus then splits cleanly:

$$\Delta TS = \underbrace{(D_w - D_c)}_{\text{distortion shrinks: } +0.36} + \underbrace{(W_w^{\text{SO}} - W_c^{\text{SO}})}_{\text{frontier rises: } +0.38} = +0.74.$$

Both pieces are positive but small. The efficiency effect of the war is genuinely second order; the first-order story is the redistribution from merchants to consumers.

Table 10: Over-penetration and the Distortion from the Social Optimum

Regime	T (market)	T^{SO} (planner)	W^{eq}	Distortion $D = W^{\mathrm{eq}} - W^{\mathrm{SO}}$
Pre-war (c)	0.43	0.073	74.15	−2.42
War (d)	0.51	0.137	74.89	−2.06

Notes. W^{eq} is welfare (34) evaluated at the market allocation; W^{SO} is its value at the social optimum (35), with $W_c^{\mathrm{SO}} = 76.57$ and $W_w^{\mathrm{SO}} = 76.95$. Yuan per unit consumer mass.

5.3 Counterfactuals

The war is both privately optimal for the platforms and mildly efficiency-improving in the aggregate. Does that leave any role for the regulator? It does, because the over-penetration distortion remains, and because the war's first-order effect is distributional. We evaluate the two policy levers regulators used: a cap on the consumer subsidy and a cap on the merchant commission.

Capping the subsidy. This is the lever Chinese regulators actually pulled. Over 2025–2026 the State Administration for Market Regulation pressed the platforms to end "involution-style" competition and, by mid-2026, had drafted binding rules—the "Ten Guidelines on Subsidy Practices"—that restrict and ultimately prohibit consumer subsidies. We simulate this policy directly. Holding the commission at its war level $r = 0.285$, we slide the subsidy down from its equilibrium value +5.92 toward the pre-war −3.50 and trace the equilibrium (Table 11). Penetration falls from 51 to 44 percent, back toward the socially optimal level, and total surplus rises monotonically, by +0.77 at the floor. But the move is overwhelmingly redistributive in the opposite direction from the war: consumers lose 8.77 while merchants gain 5.98 and platforms gain 3.56. The merchant recovery is entirely an *offline* story—the revived dine-in rent pool (+6.30) as p_{off} climbs from 64 back to 71—while the online information rent slightly shrinks.

What, then, is the effect of the Chinese subsidy ban? On our calibration it is the mirror image of the war. Because the war pushed online penetration far above the social optimum, restraining the subsidy does raise total surplus, but the gain is second order—about three-quarters of a yuan, against transfers an order of magnitude larger. The policy's first-order effect is distributional: it claws roughly nine yuan back from consumers and returns it to producers, with merchants the principal beneficiaries through their revived storefront margins. The ban is therefore best read not as an efficiency measure but as a *producer-protection* measure. It restores the offline rent pool the war destroyed and undoes the merchant losses we document in Section 3—at the direct expense of the consumers the war had benefited. Whether that trade is desirable depends on how the regulator weighs consumers against merchants, a judgment the efficiency calculus alone cannot settle.

Table 11: Capping the Subsidy (commission fixed at r = 0.285)

s	T	p_{off}	ΔCS	$\Delta\Pi^{\text{mer}}$	$\Delta\Pi^{\text{plat}}$	ΔTS
+5.92	0.510	64.4	0.00	0.00	0.00	0.00
+1.21	0.475	67.5	−4.34	+2.84	+1.93	+0.43
−3.50	0.441	70.7	−8.77	+5.98	+3.56	+0.77

Notes. Changes relative to the war baseline s = +5.92. Yuan per unit consumer mass.

The waterbed effect: capping the commission defuses the war. The more striking result concerns the commission. Suppose the regulator caps only the merchant commission and lets each platform choose its subsidy optimally. The platform's per-order take is $r\,p_{\text{on}} - s$; lowering r makes each online consumer less valuable, so the platform's own optimal subsidy falls. This waterbed effect runs in the regulator's favor. The endogenous Nash subsidy falls steadily with the commission cap—from +5.92 at r = 0.285 to −3.25 at r = 0.200—so that forcing the commission back to its pre-war level drives the platform's voluntary subsidy back to roughly the pre-war level as well. Capping the commission alone (policy B in Table 12) nearly reproduces capping both instruments (policy A): the war defuses itself, with no explicit ban on subsidies. A fee-relief rule carries an anti-subsidy effect for free.

Table 12: Regulatory Counterfactuals

Policy	r	s	T	ΔCS	$\Delta\Pi^{\text{mer}}$	$\Delta\Pi^{\text{plat}}$	ΔTS
Baseline (war)	0.285	+5.92	0.51	0	0	0	0
A. Cap both	0.200	−3.50	0.48	−4.05	+2.96	+1.78	+0.68
B. Cap commission only	0.200	$-3.25^{\dagger}$	0.48	−3.84	+2.81	+1.69	+0.66
C. Maximize total surplus	0.10	−22	0.39	−16.5	+12.3	+5.9	+1.68
D. Maximize $CS + \Pi^{\text{mer}}$	0.10	+16	0.64	+14.0	−4.8	−11.8	−2.66

Notes. Changes relative to the war baseline. † the platform's endogenous optimal subsidy at r = 0.200. Policies C and D are grid optima and are read for direction, not for precise corner values.

Consumers versus producers. The last two rows draw the policy frontier. Maximizing total surplus (policy C) points toward a low commission paired with a deep tax on online orders, which pushes penetration back toward the efficient level. Favoring the consumer-plus-merchant coalition (policy D) points the other way, toward a low commission and a large subsidy, which delights consumers but crushes both merchants and platforms. The recurring lesson is that merchants and platforms always move together and opposite to consumers. A regulator can protect consumers, or it can protect producers, but no instrument in this market protects both at once. That tension, not a free lunch in efficiency, is the central policy message of the episode.

6 Conclusion

We set out to ask who pays when platforms compete next to an offline direct channel, and whether that competition still raises welfare once the offline market it cannibalizes is counted. China's 2025 food-delivery subsidy war supplies an unusually clean answer. The war did exactly what subsidies are designed to do on the delivery channel—prices fell and orders rose—yet merchant revenue fell. Order volume and revenue moved apart because subsidized delivery pulled customers off the high-margin dine-in counter. Restaurants handled more traffic and took home less money.

The model explains the paradox and turns it into a welfare statement. The merchant's outside option is not fixed; it is the storefront, whose markup erodes as the platforms pull consumers online. A subsidy paid to consumers is therefore, in its incidence, a tax on the merchant's outside option. Modeling the war as a move from a captive duopoly, in which consumers never compare platforms, to a contestable one, in which they do, we showed that the shift unconditionally raises penetration and subsidies, expands the merchant base, compresses platform profit into a prisoner's dilemma, helps consumers, and hurts merchants. The harm is off the books: it runs through the offline markup, a quantity no platform contract reports, while the commission and the posted online price can move the other way.

The calibration sized these flows. The war is overwhelmingly a redistribution—about nine yuan per unit of demand from merchants and platforms to consumers—with only a small, second-order effect on total surplus. The policy counterfactuals carry the sharpest practical lesson. Because the platform extracts less from each online order once its commission is capped, a cap on the merchant commission lowers the platform's own optimal subsidy and defuses the war without any direct ban; fee relief and subsidy restraint turn out to be the same policy. But the regulator faces a real constraint: merchants and platforms always move together and opposite to consumers, so no instrument protects sellers and shoppers at the same time.

The broader implication reaches past food delivery. Wherever a platform competes beside a direct channel that sellers also use—hotels and direct booking, marketplaces and own storefronts, ride hailing and the street hail—the welfare case for platform competition cannot be read from prices and variety inside the platform alone. The decisive margin is what competition does to the value of the seller's outside option. When that option is endogenous, vigorous platform competition can lower consumer prices and still leave the sellers who supply the market worse off. Reading competition's incidence correctly requires looking at the channel the platform displaces, not only the one it runs.

A Supplemental Empirical Materials

The average baseline and mechanism estimates above mask substantial heterogeneity, and that heterogeneity is itself informative about the economics of the war. We cut the sample along four dimensions: a merchant's subsidy participation, its organizational form, its reliance on dine-in versus delivery, and its product category. Throughout, we re-estimate the baseline equation (1) on each subsample, using PPML for count and revenue outcomes and OLS for log delivery price. The recurring theme is that the traffic–revenue paradox is pervasive: it appears in essentially every subgroup, including those that chose *not* to participate in the subsidies.

A.1 Subsidy Participation: A Prisoner's Dilemma

We first split merchants by the intensity of their delivery-subsidy participation during the war into high-participation, partial-participation, and non-participating groups. We stress at the outset that this partition is *descriptive*, not causal: participation is measured from behavior during the treatment window and is therefore an equilibrium choice, a "bad control" in the sense of Angrist and Pischke (2009), so the across-group differences confound the war's effect with selection. With that caveat, the pattern in Figure 2 is striking precisely because it is uniform.

Every group loses revenue. High-participation merchants flood with delivery orders (+26.8%) bought through a steep cut in the delivery price (−24.2%), yet their total actual revenue still falls by 7.0%. Non-participants tell the opposite story with the same ending: they do not cut prices, their delivery price in fact rises slightly (+2.8%), and they shed delivery orders (−18.0%), but their total actual revenue nonetheless declines by 4.7%. Merchants who participate aggressively lose money by discounting; merchants who abstain lose money because subsidized competitors siphon away their customers. The configuration is a prisoner's dilemma: participation is individually rational given rivals' behavior, but the symmetric outcome leaves the entire sector worse off.

A.2 Organizational Form, Business Structure, and Product Category

Table 13 turns to structural merchant characteristics, which, unlike subsidy participation, are largely predetermined.

Independent versus chain (Panel A). Chains weather the war markedly better than independent single-store merchants. Chains convert the order boom into essentially flat revenue (+0.5%, not economically distinguishable from zero), whereas independents suffer a significant revenue decline (−1.8%) despite also gaining orders. Scale and bargaining power appear to cushion the revenue incidence of the subsidies, a Matthew effect within the restaurant sector.

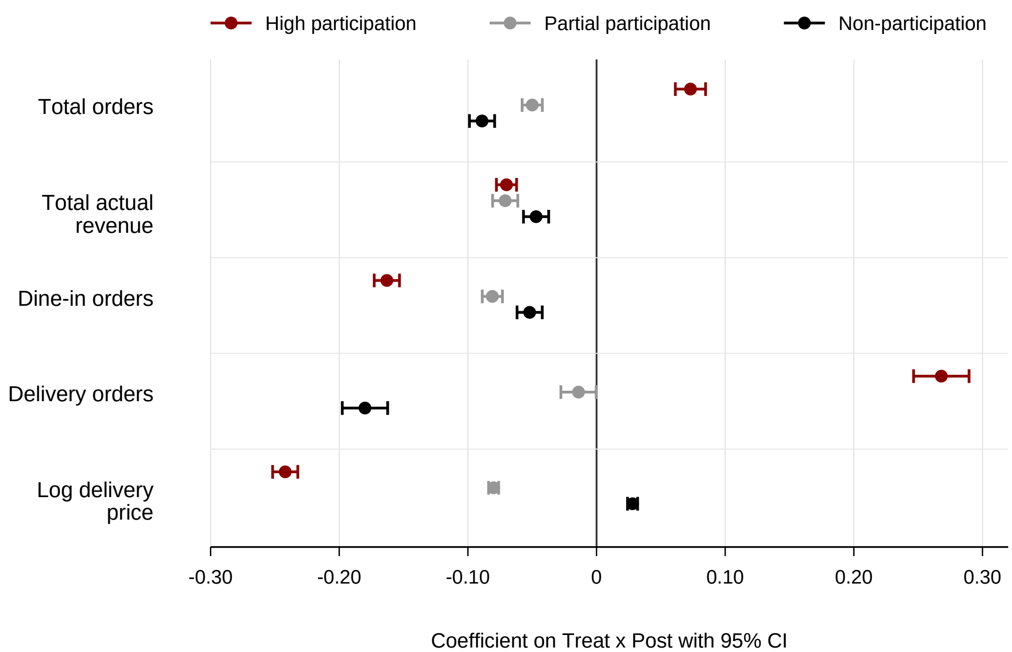


Figure 2: Heterogeneity by Subsidy Participation

Notes. Points report the coefficient on Post $\times$ Treat from separate regressions of equation (1) by subsidy-participation group; horizontal bars are 95% confidence intervals. The first four outcomes are estimated by PPML and log delivery price by OLS. Subsidy participation is measured from treatment-period behavior and is endogenous; these contrasts are descriptive, not causal. Fixed effects, controls, and clustering are as in Table 2.

Dine-in orientation (Panel B). Splitting by the dine-in share of revenue sharpens the channel-substitution mechanism. Delivery-oriented merchants (low dine-in share) bear the largest revenue loss in the entire analysis (−10.3%), accompanied by the steepest dine-in contraction (−18.5%). Dine-in-oriented merchants are comparatively insulated (−4.9%): a heavy dine-in business is a partial moat against a price war fought on the delivery channel.

Product category (Panel C). Exposure also varies by cuisine. Delivery-oriented categories experience the most violent price war, delivery orders surge 30.8% on a 24.2% delivery price cut, but it is full-service Chinese restaurants that absorb the deepest total revenue decline (−8.9%), as their high-margin dine-in trade (−13.0% in orders) is the most valuable to lose. The disruption is thus concentrated where substitutability is highest, but the revenue damage is concentrated where dine-in margins are richest.

Table 13: Heterogeneity by Merchant Characteristics

	(1) Total orders	(2) Total actual revenue	(3) Dine-in orders	(4) Delivery orders	(5) Delivery price (log)
Panel A. Organizational form					
Independent (single-store)	0.033***	−0.018***	−0.044***	0.134***	−0.112***
	(0.003)	(0.002)	(0.003)	(0.007)	(0.003)
Chain	0.114***	0.005**	−0.073***	0.289***	−0.184***
	(0.005)	(0.003)	(0.004)	(0.010)	(0.004)
Panel B. Dine-in orientation					
High dine-in share	−0.001	−0.049***	−0.081***	0.175***	−0.126***
	(0.003)	(0.004)	(0.004)	(0.011)	(0.004)
Low dine-in share	0.032***	−0.103***	−0.185***	0.151***	−0.175***
	(0.007)	(0.007)	(0.006)	(0.011)	(0.004)
Panel C. Product category					
Delivery-oriented cuisine	0.112***	−0.068***	−0.179***	0.308***	−0.242***
	(0.011)	(0.010)	(0.010)	(0.015)	(0.006)
Chinese full-service	−0.007	−0.089***	−0.130***	0.126***	−0.152***
	(0.007)	(0.006)	(0.005)	(0.014)	(0.003)
Other	0.002	−0.057***	−0.075***	0.090***	−0.101***
	(0.004)	(0.006)	(0.005)	(0.010)	(0.003)

Notes. Each cell is the coefficient on Post × Treat from a separate PPML (columns 1–4) or OLS (column 5) regression on the indicated subsample. Splits are on predetermined merchant attributes. Fixed effects and controls as in Table 2; standard errors clustered at the city × calendar-month level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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